Rohit Das

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CAREER OBJECTIVE

To contribute to cutting-edge research and development in VLSI and secure hardware design, with the goal of driving innovation in semiconductor technology. M.Tech student specializing in SoC Design with hands-on experience in Digital VLSI, STA, Physical Design, and FPGA implementation using Synopsys and Vivado tools.

EDUCATION

Degree/Certificate	**Institute/Board**	**CGPA / %**	**Year**
M.Tech (SoC Design)	IIT Palakkad	7.06	2025
B.Tech (IT)	St. Thomas' College of Engineering	8.84	2024
Class XII	Siliguri Baradakanta Vidyapith	83%	2020
Class X	New St John's High School	80%	2018

TECHNICAL SKILLS

- **Digital IC Design:** RTL Design, Static Timing Analysis (STA), Physical Design Flow, Low Power Techniques,Computer Architecture.
- **EDA Tools:** Synopsys VCS, Synopsys Design Compiler, IC Compiler II, Vitis HLS, Xilinx Vivado.
- **Programming:** Verilog HDL, Python, Java, Data Structures & Algorithms.

PROJECTS

16-bit ALU: RTL to GDSII Physical Design Flow *Academic Project*

Designed a 16-bit ALU in Verilog supporting 16 operations and performed the full Physical Design flow. Executed Floorplanning, Placement, and Clock Tree Synthesis (CTS) using Synopsys IC Compiler II. Performed routing and timing closure, ensuring the design met setup/hold constraints with zero Slack violations.

Implementation of CORDIC Algorithm *Academic Project*

Designed and implemented the CORDIC algorithm in Verilog supporting both rotation and vectoring modes. Developed an iterative architecture featuring an FSM-based controller with handshaking and a hardware-efficient barrel shifter. Additionally, implemented a 16-stage pipelined architecture to maximize throughput. Performed a comparative analysis of both designs regarding hardware resource utilization, latency, and throughput.

Design and Implementation of 4-Tap FIR Filter *Academic Project*

Developed an 4-tap digital FIR filter using Direct Form architecture in Verilog. Optimized the hardware using an adder tree structure and validated signal processing accuracy and group delay through comprehensive simulation in Xilinx Vivado.

FSM-Based Traffic Light Controller & Synchronous Counter *Academic Project*

Implemented a Traffic Light Controller using Finite State Machine (FSM) abstraction in Verilog, incorporating specific timing cycles for highway and crossroad transitions. Verified sequential logic through RTL simulation.

Trojan Detection using Cellular Automata Apr 2023 – May 2024

Formulated a novel approach to detect and prevent hardware Trojans in IoT devices by utilizing Cellular Automata (CA) and Physical Unclonable Functions (PUFs) as implicit detection mechanisms.

ACHIEVEMENTS

- **Conference Paper:** "A New Technique of Cellular-Automata-Based PUF Design in Nanoscale Crossbar Circuits"
  Link to Publication